

Name _____

Tie Breaker: Points scored on Stated and Geometry Problems

+ _____ + _____ + _____

5x (Last Problem Attempted) + _____ + _____ + _____

7x (Number Incorrect) - _____ - _____ - _____

2x (Number Incorrect SDs) - _____ - _____ - _____

TOTAL SCORE _____

Grader Initials _____

UIL Calculator Applications

Test 26A (Invitational A)

DO NOT OPEN THE TEST UNTIL INSTRUCTED TO BEGIN

- I. Calculator Applications rules and scoring—See UIL Constitution
II. How to write the answers

A. For all problems except stated problems as noted below—write three significant digits.

1. Examples (* means correct but not recommended)

Correct: 12.3, 123, 123.*, 1.23x10*, 1.23x10⁰*

1.23x10¹, 1.23x10⁰¹, .0190, 0.0190, 1.90x10⁻²

Incorrect: 12.30, 123.0, 1.23(10)², 1.23·10², 1.230x10²,
1.23*10², 0.19, 1.9x10⁻², 19.0x10⁻³, 1.90E-02

2. Plus or minus one digit error in the third significant digit is permitted.

B. For stated problems

1. Except for integer, dollar sign, and significant digit problems, as detailed below, answers to stated problems should be written with three significant digits.

2. Integer problems are indicated by (integer) in the answer blank. Integer problems answers must be exact, no plus or minus one digit, no decimal point or scientific notation.

3. Dollar sign (\$) problems should be answered to the exact cent, but plus or minus one cent error is permitted. Answers must be in fixed notation. The decimal point and cents are required for exact-dollar answers.

4. Significant digit problems are indicated by underlined numbers and by (SD) in the answer blank. See the UIL Constitution and Contest Manual for details.

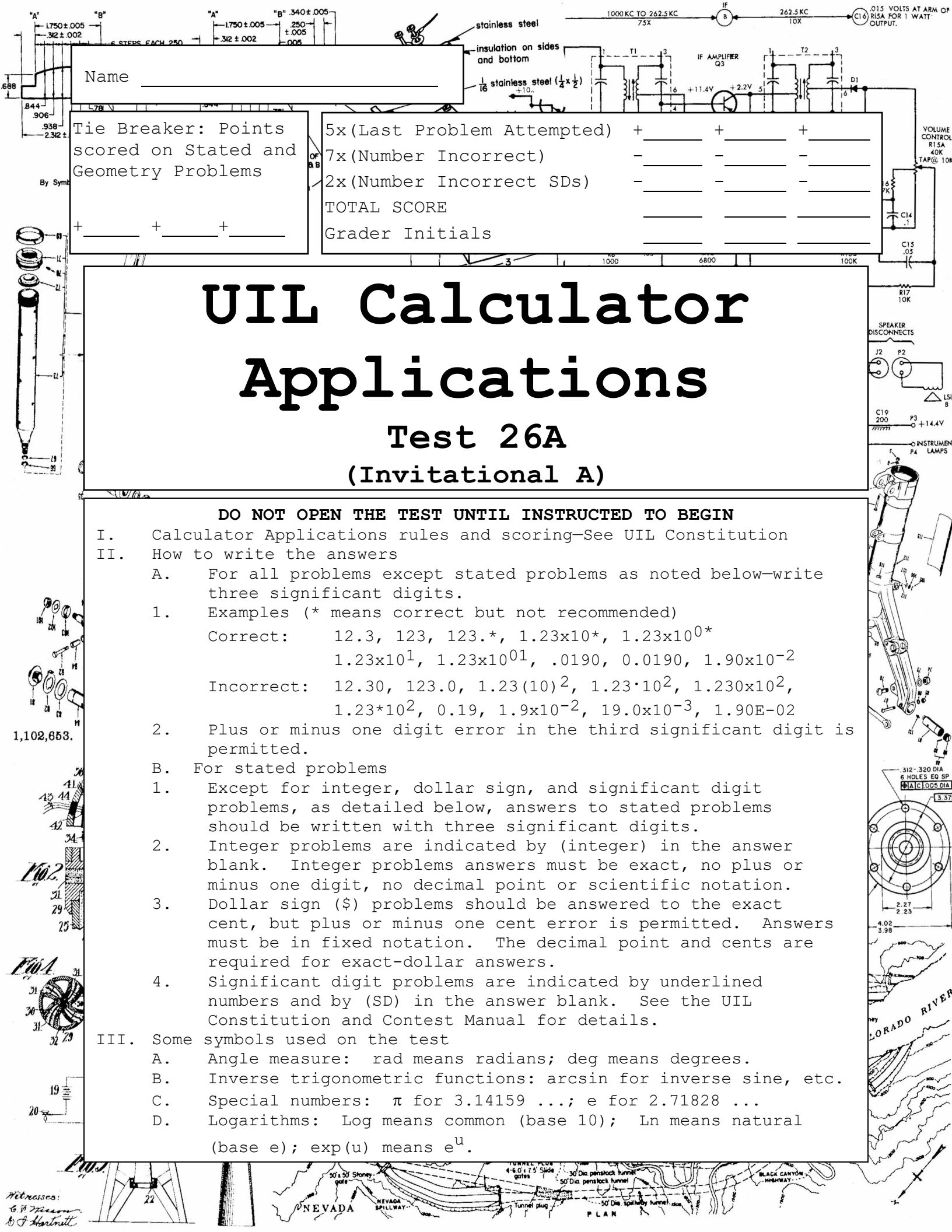
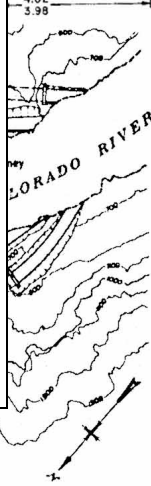
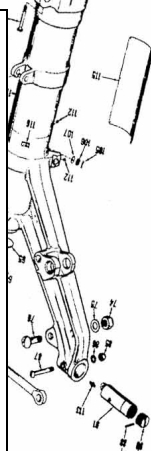
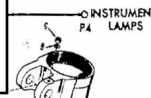
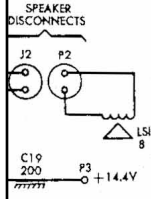
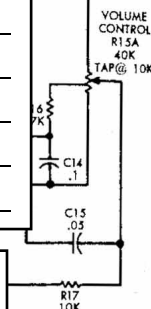
III. Some symbols used on the test

A. Angle measure: rad means radians; deg means degrees.

B. Inverse trigonometric functions: arcsin for inverse sine, etc.

C. Special numbers: π for 3.14159 ...; e for 2.71828 ...

D. Logarithms: Log means common (base 10); Ln means natural (base e); exp(u) means e^u.



26A-1. $(-24.7 - 20.7)/(-38.2)$ ----- 1=_____

26A-2. $-0.918/0.575 + 0.744 - 1.6$ ----- 2=_____

26A-3. $\frac{(2.62)(21.2)(71.9)}{-76} + 8.75$ ----- 3=_____

26A-4. $\frac{(2520 - 2040)}{\{(-0.191)/(-0.706)\}} + (449 - 263)$ ----- 4=_____

26A-5. $\frac{49900 + 43300}{(0.00745)(0.0512)(0.0186)} + 1.33 \times 10^{10} - 1.22 \times 10^{10}$ ----- 5=_____

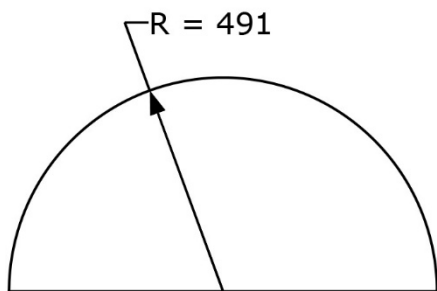
26A-6. What is 430 divided by 42.3? ----- 6=_____

26A-7. What is twice the product of 32.1 and 7.66? ----- 7=_____

26A-8. What is the sum of π^e and e^π ? ----- 8=_____

26A-9.

SEMICIRCLE

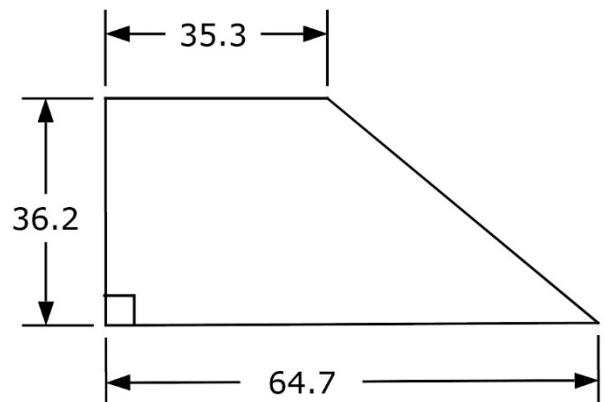


Area = ?

26A-9 = _____

26A-10.

RIGHT TRAPEZOID



Perimeter = ?

26A-10 = _____

26A-11. $\frac{(-20.5)(5660) + (-312)(419)}{-1.43 + 0.255 - (-1.1)(0.977)}$ ----- 11=_____

26A-12. $\frac{-0.0803(4.80 \times 10^{-5} + 2.12 \times 10^{-5})}{(950 - 998)(0.0892)} - \frac{-7.01 \times 10^{-7}}{0.28 - 0.097}$ ----- 12=_____

26A-13. $\frac{(-0.656)(434 - 279)\{-0.0774 - (-0.268)(0.0533)\}}{(0.418 + 0.0702)(-0.918 - 1.23)}$ ----- 13=_____

26A-14. $\frac{(4710 + 2870 - 2410)(0.0098 + 0.0102 - 0.00563)}{(0.455 - 0.21)(-0.861)(-0.524 - 0.13)}$ ----- 14=_____

26A-15. $\frac{56900 + 1.50 \times 10^5 - (25100 + 89000)(\pi - 0.292)}{(-669)(21.3)(-17.5)(874 - 797 + 976)}$ ----- 15=_____

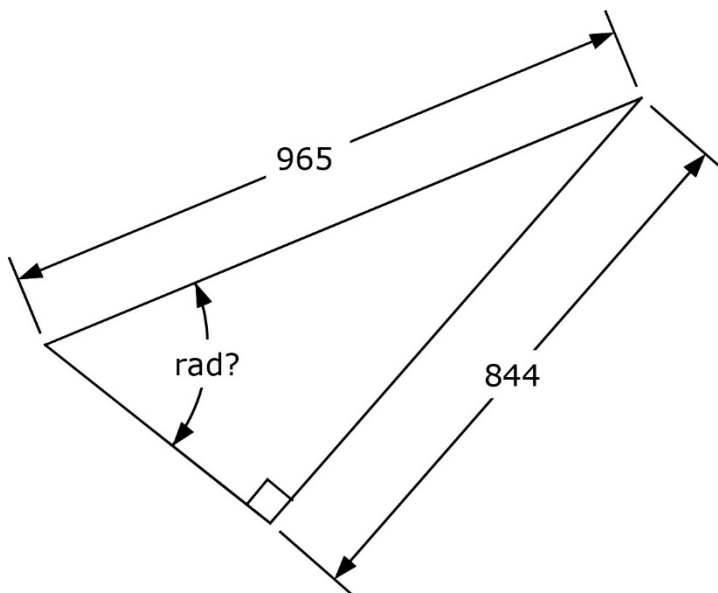
26A-16. A bag of plain M&M candies costs \$1.29 and contains 104 candies. What is the cost of a single M&M candy? ----- 16=_____cents

26A-17. Mandy eats out. Her meal cost is \$18.45. Tax is 8.125%, and she adds a tip of 18% of the cost of the meal and tax. What is her total? ---- 17=\$_____

26A-18. A cheap Taylor Swift concert ticket is \$689. What is the percent increase in ticket cost for an expensive ticket, \$1500? ----- 18=_____%

26A-19.

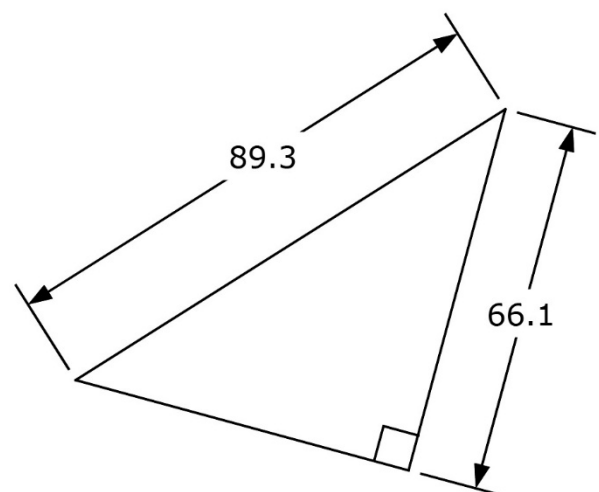
RIGHT TRIANGLE



26A-19 = _____

26A-20.

RIGHT TRIANGLE



Area = ?

26A-20 = _____

26A-21. $\left[\frac{(0.978)(0.707)}{1.79} + 0.207 \right]^2 + \sqrt{0.0567}$ ----- 21=_____

26A-22. $\frac{1}{\pi + 9.29} + \frac{1}{0.108 - 0.134} + \frac{1}{(0.101)}$ ----- 22=_____

26A-23. $\frac{\sqrt{0.71 + 0.487 + (1.2)/(3.87)}}{-9.85 + 9.67}$ ----- 23=_____

26A-24. $[-31.2 + \sqrt{862}]^2 \times [705 + 1650]^2 \times \sqrt{0.0032/0.0324}$ ----- 24=_____

26A-25. $(\pi)(0.0158)\sqrt{(-0.726)^2/0.958} + 1/\sqrt{20.1 + 36.3}$ ----- 25=_____

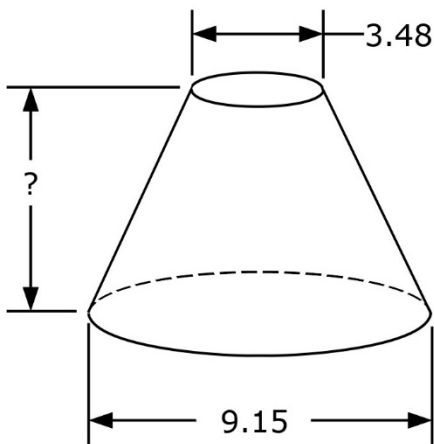
26A-26. Stan wants to build a cylindrical cistern that holds 700 gallons of water. How tall must the cistern be if its diameter is 6 ft? -----26=_____ ft

26A-27. Ancient Jericho occupied 40,000 square meters and is assumed to be circular. Joshua and his troops marched around the city seven times before it fell. How far did they travel? -----27=_____ mi

26A-28. Michaelangelo's statue *David* weighs 12,478.16 lbs and has an average cross section of 4.633 ft². If marble density is 160 lbs/ft³, how tall is the statue? -----28=_____ ft(SD)

26A-29.

FRUSTUM

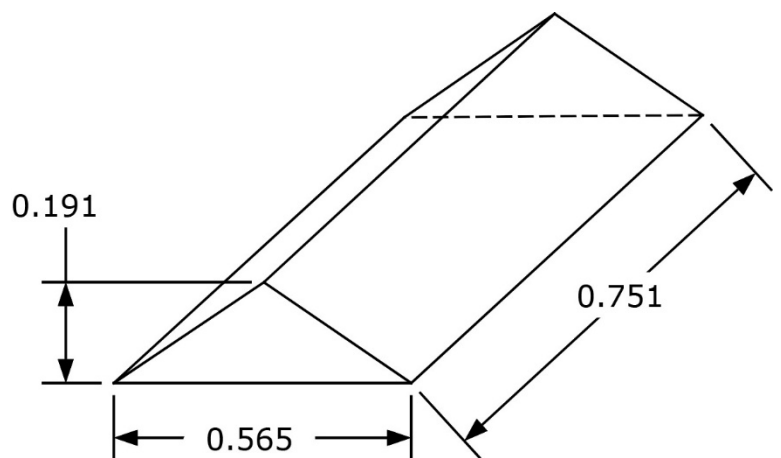


Volume = 198

26A-29 = _____

26A-30.

ISOSCELES TRIANGLE PRISM



Total Surface Area = ?

26A-30 = _____

$$26A-31. \sqrt{\frac{2.11}{\sqrt{43.1 + 29.3}}} \times \left[\frac{1}{(5.42 - 4.52)^2} + \frac{1}{(\pi + 0.294)^2} \right] \text{ ----- } 31 = \underline{\hspace{2cm}}$$

$$26A-32. \left[\frac{-79.6}{-77.1 + 30.7} + \pi \right] \times \left\{ 2720 + (-72.4)^2 - \sqrt{8.09 \times 10^7} \right\} \text{ ----- } 32 = \underline{\hspace{2cm}}$$

$$26A-33. \frac{\sqrt{(0.00519)/\{(0.00285)/\sqrt{0.00638}\}}}{0.125 + (0.653)(1.8)} + \{0.0688 + 0.553\}^{1/2} \text{ ----- } 33 = \underline{\hspace{2cm}}$$

$$26A-34. \frac{(4.82 \times 10^5)^2 (8.96 \times 10^{-13} + 8.45 \times 10^{-13})}{40.7 + (-0.243)(792)} + \frac{1}{\frac{1}{-0.0016} + \frac{1}{(5.72 \times 10^{-4})}} \text{ ----- } 34 = \underline{\hspace{2cm}}$$

$$26A-35. \frac{\frac{1}{2090} + \frac{260}{(518 + 486)^2} - \frac{\sqrt{0.152}}{(-38.9)^2}}{(-494 + 1020)^2 + (-3.83 \times 10^5)} \text{ ----- } 35 = \underline{\hspace{2cm}}$$

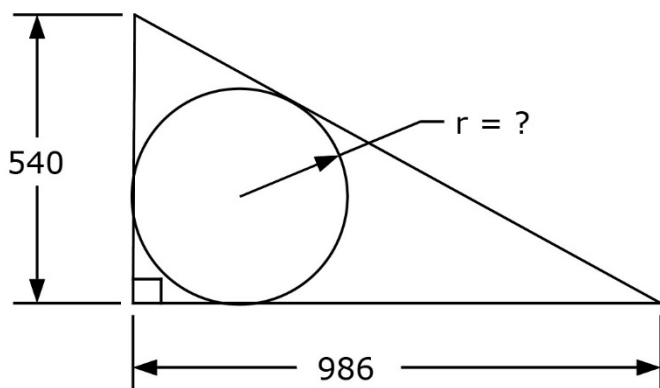
26A-36. Mt. Everest peak is at an elevation of 29,032 ft. How far could a climber see from the peak on a clear day if his view was unobstructed? -----36= mi

26A-37. The playing area of a football field is 48,000 ft². What is the distance of one lap around its periphery? -----37= ft

26A-38. Brenda weighs 145 lbs and goes backpacking. Her recommended backpack weight is 20% of her body weight. If she carries 1.5 gal of water, how much does all the other backpack equipment weigh? -----38= lbs

26A-39.

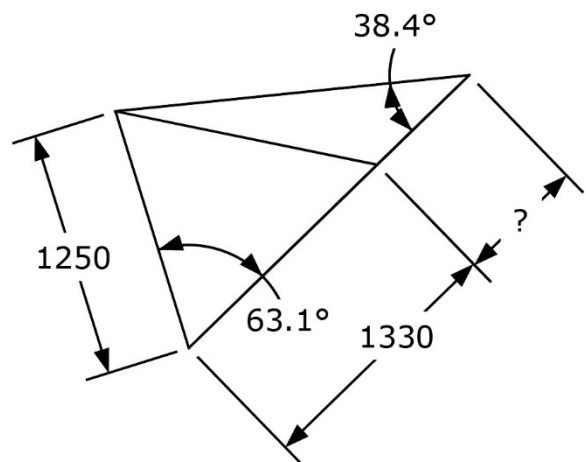
INSCRIBED RIGHT TRIANGLE



26A-39 =

26A-40.

SCALENE TRIANGLES



26A-40 =

26A-41. $(0.00659)(0.0064)10^{\{-0.00101/-8.66 \times 10^{-4}\}}$ ----- 41=_____

26A-42. $\frac{e^{+0.424} + e^{-0.695}}{(-85700 + 31400)}$ ----- 42=_____

26A-43. $\frac{(-4.85)\text{Log}(5.07 - 1.89)}{(-9.16)}$ ----- 43=_____

26A-44. $(0.0786 + 0.165)^{-(0.446 + 0.896)}$ ----- 44=_____

26A-45.(deg) $\{(3.94)\sin(-66.3^\circ)\} \times \{(1.36)\cos(-97.8^\circ)\}$ ----- 45=_____

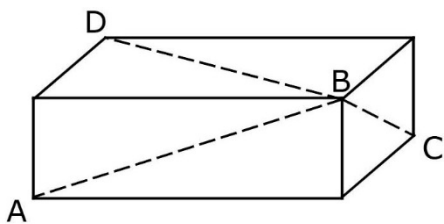
26A-46. A large truck holds 8500 7-in long cantaloupes. How many trucks would be needed to hold 12,000 15-in long watermelons? -----46=_____ integer

26A-47. The second through sixth Fibonacci numbers are (2, 1), (3, 2), (4, 3), (5, 5), and (6, 8). What is the slope of the best-fit line through these data? -----47=_____

26A-48. (rad) What is t if $10 + \sin(\sqrt{t}) = 3t$? -----48=_____

26A-49.

CUBOID



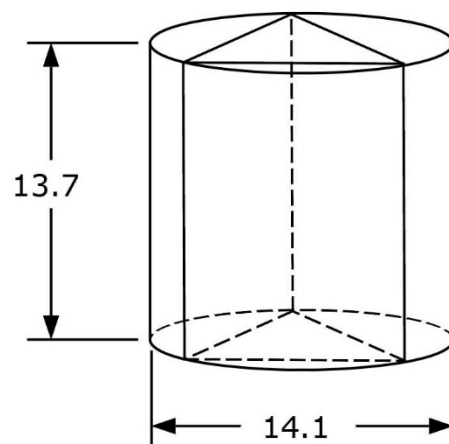
$BD = 21.5 \quad AB = 20.8 \quad BC = 10.6$

Volume = ?

26A-49 = _____

26A-50.

EQUILATERAL TRIANGLE PRISM AND CYLINDER



Volume(Cylinder) - Volume(Prism) = ?

26A-50 = _____

26A-51. $10^{+(0.791)} + 10^{-(0.211)} + [10^{(0.457/0.699)} - 10^{(0.509)}]^{1/2}$ 51=_____

26A-52. $\frac{28 + e^{(2.46 + 1.13)}}{0.0952 - e^{-(0.937 - 0.416)}}$ ----- 52=_____

26A-53. $\frac{\text{Ln}\{(8290)(9840)(5650)\}}{-10700 + (-4450) \text{Ln}(7250)}$ ----- 53=_____

26A-54. $\frac{1}{(0.697)^{(-0.134)}} + (0.912 + 0.488)^{(0.319 - 0.19)}$ ----- 54=_____

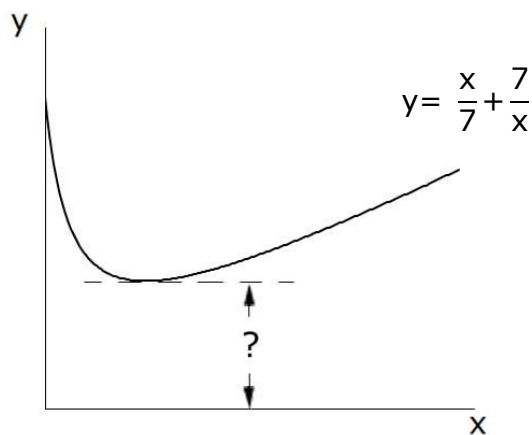
26A-55.(rad) $\frac{\arctan\{1.98 + (4.42)(0.232)\}}{\arcsin\{(0.0835 + 0.0368)/0.276\}}$ ----- 55=_____

26A-56. What is the slope of the curve $y = x^3(x-9)^2$ when x equals -5? -----56=_____

26A-57. A hollow cylinder with end caps holds a constant volume of liquid. Material for the circular cylinder end caps costs \$2/in², and material for the cylinder side walls costs \$0.50/in². What is the height to diameter ratio that minimizes the cylinder cost? -----57=_____

26A-58. Calculate J₃₃ if $\mathbf{J} = 3\mathbf{K} + 4\mathbf{L}$, $\mathbf{K} = \begin{bmatrix} 2 & 8 & 8 \\ 8 & 7 & -2 \\ 8 & -2 & 6 \end{bmatrix}$ and $\mathbf{L} = \begin{bmatrix} 5 & 2 & 9 \\ 2 & -6 & 1 \\ 9 & 1 & 5 \end{bmatrix}$. 58=_____

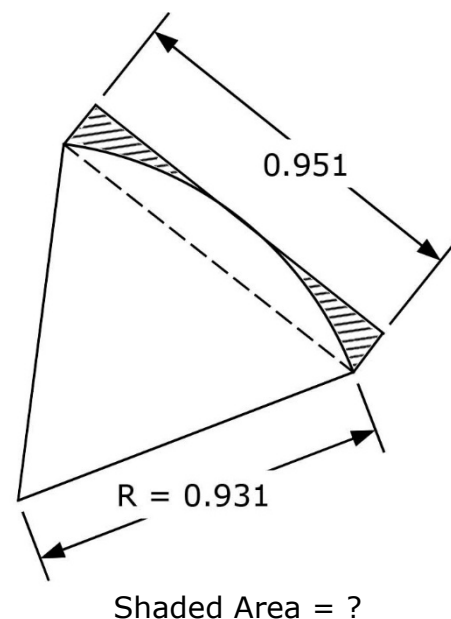
26A-59.



26A-59 = _____

26A-60.

ISOSCELES TRIANGLE, SEGMENT, AND RECTANGLE



26A-60 = _____

26A-61. Tom and Alan start a race around a 440-yd oval track. Tom runs a 6.5 min mile, and Alan runs slower. Tom "laps" Alan in 4 min 38.6 s, passing him. How long does it take Alan to run a mile? -----61=_____min

26A-62. The time since the Big Bang is estimated to be 4.36×10^{17} s. What is this number raised to the power 500? -----62=_____

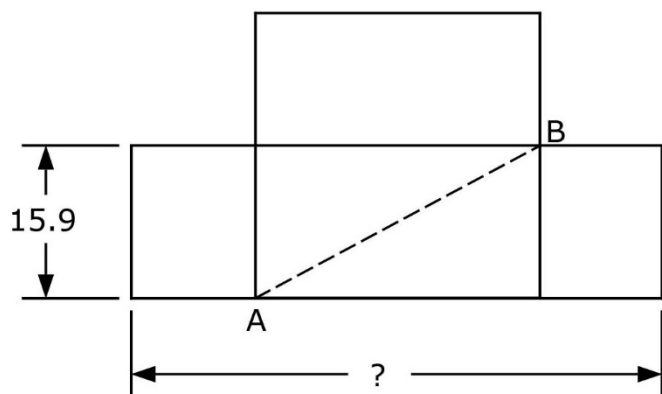
26A-63. Jacqueline bikes at 20 mph. At a crosswalk, she drops a rock from a 6.5 ft height. How far away from the crosswalk does the rock hit the ground? -----63=_____ft

26A-64.

SQUARE AND RECTANGLE

$$AB = 33.8$$

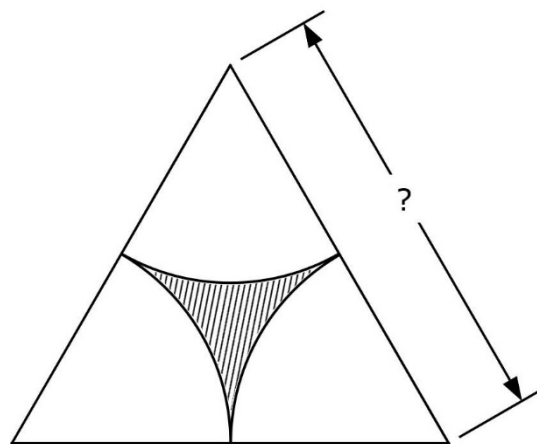
$$\text{Area}(\text{Square}) = \text{Area}(\text{Rectangle})$$



26A-64 = _____

26A-65.

EQUILATERAL TRIANGLE AND IDENTICAL SECTORS



$$\text{Shaded Area} = 6800$$

26A-65 = _____

26A-66. $\ln \left[\frac{(6.21)^2 - 2(6.21)(13.4) + (13.4)^2}{(86)^2} \right]^2$ ----- 66=_____

26A-67. (rad) $\cos(1.76 - 0.386) - \cos(1.76 + 0.386)$ ----- 67=_____

26A-68. (deg) $\{ \cos^2(89.9^\circ) - \sin^2(89.9^\circ) \} \times \frac{\tan(89.9^\circ)}{1 - \tan^2(89.9^\circ)}$ ----- 68=_____

26A-69. $(0.579) - \frac{(0.579)^2}{2} + \frac{(0.579)^3}{3} - \frac{(0.579)^4}{4}$ ----- 69=_____

26A-70. (rad) $\frac{(0.874)(1.3) - \ln \{ (1.63) + (-5.27)e^{(-2.79)} \}}{\arcsin \{ (0.0196)/(0.0644 + 0.0294) \}}$ ----- 70=_____

26A-1	= 1.19 = 1.19×10^0	26A-11	= 2.46×10^6	26A-21	= 0.590 = 5.90×10^{-1}
26A-2	= -2.45 = -2.45×10^0	26A-12	= 5.13×10^{-6}	26A-22	= -28.5 = -2.85×10^1
26A-3	= -43.8 = -4.38×10^1	26A-13	= -6.12 = -6.12×10^0	26A-23	= -6.82 = -6.82×10^0
26A-4	= 1960 = 1.96×10^3	26A-14	= 539 = 5.39×10^2	26A-24	= 5.90×10^6
26A-5	= 1.42×10^{10}	26A-15	= -0.000450 = -4.50×10^{-4}	26A-25	= 0.170 = 1.70×10^{-1}
26A-6	= 10.2 = 1.02×10^1	26A-16	= 1.24 = 1.24×10^0	26A-26	= 3.31 = 3.31×10^0
26A-7	= 492 = 4.92×10^2	26A-17	= \$23.54	26A-27	= 3.08 = 3.08×10^0
26A-8	= 45.6 = 4.56×10^1	26A-18	= 118 = 1.18×10^2	26A-28	= 17 = 1.7×10^1 (2SD)
26A-9	= 379,000 = 3.79×10^5	26A-19	= 1.06 = 1.06×10^0	26A-29	= 5.92 = 5.92×10^0
26A-10	= 183 = 1.83×10^2	26A-20	= 1980 = 1.98×10^3	26A-30	= 1.04 = 1.04×10^0

26A-31	= 0.657 = 6.57x10 ⁻¹	26A-41	= 0.000619 = 6.19x10 ⁻⁴	26A-51	= 7.93 = 7.93x10 ⁰	26A-61	= 10.0 = 1.00x10 ¹
26A-32	= -5020 = -5.02x10 ³	26A-42	= -3.73x10 ⁻⁵	26A-52	= -129 = -1.29x10 ²	26A-62	= 5.54x10 ⁸⁸¹⁹
26A-33	= 1.08 = 1.08x10 ⁰	26A-43	= 0.266 = 2.66x10 ⁻¹	26A-53	= -0.000534 = -5.34x10 ⁻⁴	26A-63	= 18.6 = 1.86x10 ¹
26A-34	= -0.00178 = -1.78x10 ⁻³	26A-44	= 6.65 = 6.65x10 ⁰	26A-54	= 2.00 = 2.00x10 ⁰	26A-64	= 56.0 = 5.60x10 ¹
26A-35	= -4.50x10 ⁻⁹	26A-45	= 0.666 = 6.66x10 ⁻¹	26A-55	= 2.77 = 2.77x10 ⁰	26A-65	= 411 = 4.11x10 ²
26A-36	= 209 = 2.09x10 ²	26A-46	= 14 integer	26A-56	= 18200 = 1.82x10 ⁴	26A-66	= -9.93 = -9.93x10 ⁰
26A-37	= 920 = 9.20x10 ²	26A-47	= 1.70 = 1.70x10 ⁰	26A-57	= 4.00 = 4.00x10 ⁰	26A-67	= 0.740 = 7.40x10 ⁻¹
26A-38	= 16.5 = 1.65x10 ¹	26A-48	= 3.65 = 3.65x10 ⁰	26A-58	= 38.0 = 3.80x10 ¹	26A-68	= 0.00174 = 1.74x10 ⁻³
26A-39	= 201 = 2.01x10 ²	26A-49	= 1070 = 1.07x10 ³	26A-59	= 2.00 = 2.00x10 ⁰	26A-69	= 0.448 = 4.48x10 ⁻¹
26A-40	= 642 = 6.42x10 ²	26A-50	= 1250 = 1.25x10 ³	26A-60	= 0.0402 = 4.02x10 ⁻²	26A-70	= 4.13 = 4.13x10 ⁰